

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE – PILANI, HYDERABAD  
CAMPUS  
EEE/INSTR F342: POWER ELECTRONICS, SEM-II, 2025-26

TUTORIAL- 4

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**Controlled Rectifiers (Phase Controlled, Converter)**

Q1. A single-phase half wave fully controlled AC to DC converter fed by a transformer whose secondary voltage is  $400 \sin \omega t$ . Find the average load voltage, RMS load voltage, average load current and RMS load current, if the thyristor is fired at  $30^\circ$  in each positive half cycle. The load is a pure resistive load of  $50\Omega$ .

Q2. A single-phase half wave phase-controlled rectifier feeds a  $15\Omega$  resistance. The supply voltage is 230V, 50Hz and firing angle is  $90^\circ$ . Find (a) average load voltage, RMS load voltage, average load current and RMS load current, (b) Rectification efficiency (c) form factor, ripple factor (d) VA rating of the transformer feeding the rectifier (e) TUF (f) PIV across the thyristor.

Q3. A single-phase half wave rectifier using a thyristor is operating on 230V, 50Hz source to supply a lamp load. If the peak value of output voltage is 229V, what is the firing angle?

Q4. A single phase fully controlled full wave rectifier circuit uses a centre tapped transformer whose output voltage from centre tapping to outside terminal has an RMS value of 150V. It has a load resistance of  $30\Omega$ . If the firing angle is  $45^\circ$ , Derive the expression for average voltage and RMS voltage for the output voltage. Find DC voltage and current, RMS load voltage and current.

Q5. A heater of rated power 1 kW, and rated voltage 230V is controlled by a single-phase fully controlled bridge converter for an input voltage of 230V, 50Hz. If the load is to be delivered a power of 800 W, find the firing angle at which it is operating. Also calculate the DC voltage and current, RMS output voltage and current.